

The plant used in this project is the same armature-controlled DC motor from Project 02. The transfer function relating the input voltage $V(s)$ to the output angular velocity $\omega(s)$ is given by:

$$G(s) = \frac{\omega(s)}{V(s)} = \frac{K_t}{JLs^2 + (JR + bL)s + (bR + K_tK_e)}$$

Nominal Parameters

The following nominal parameter values were used throughout the base-case analysis:

Table 1 Nominal Plant Parameters

Parameter	Symbol	Value	Unit
Moment of Inertia	J	0.01	kg·m ²
Viscous Friction	b	0.1	N·m·s/rad
Armature Resistance	R	1	Ω
Armature Inductance	L	0.5	H
Torque Constant	K _t	0.01	N·m/A
Back-EMF Constant	K _e	0.01	V·s/rad

With the nominal parameters substituted, the transfer function becomes:

$$G(s) = \frac{0.01}{0.005s^2 + 0.06s + 0.1001}$$